

MODULE 3

A I R

Conservation of Natural Resources

Course Code: BCV755B

Terra Study Materials
Department of Civil Engineering



Complete Study Notes

Air & Atmosphere • Air Pollution • Pollutant Classification • NAAQS • Air Quality Index • Health & Economic Effects • Control Equipment • Ozone Depletion • Photochemical Smog

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1. Introduction to Air

1.1 What is Air?

- Air is a **non-homogeneous mixture** of different gases present in the atmosphere.
- The “composition of air” refers to the **chemical composition of the troposphere**.

1.2 The Troposphere

- The **troposphere** is the **lowest layer of the atmosphere**.
- It extends from the Earth’s surface up to about **12 km (39,000 ft) in altitude**.
- It contains about **80% of the mass** of the Earth’s atmosphere.
- **Nearly all of the atmosphere’s water vapour** exists in this layer.
- **Just three gases make up 99% of dry air: Nitrogen (N₂), Oxygen (O₂), and Argon (Ar).**

1.3 Composition of Air (by Volume)

Gas	Formula	Percentage (%)
Nitrogen	N ₂	78.084
Oxygen	O ₂	20.946
Argon	Ar	0.9340
Carbon Dioxide	CO ₂	~0.04
Neon	Ne	0.001818
Helium	He	0.000524
Methane	CH ₄	0.000187
Water Vapour	H ₂ O	0% to 5% (avg. ~0.4%)

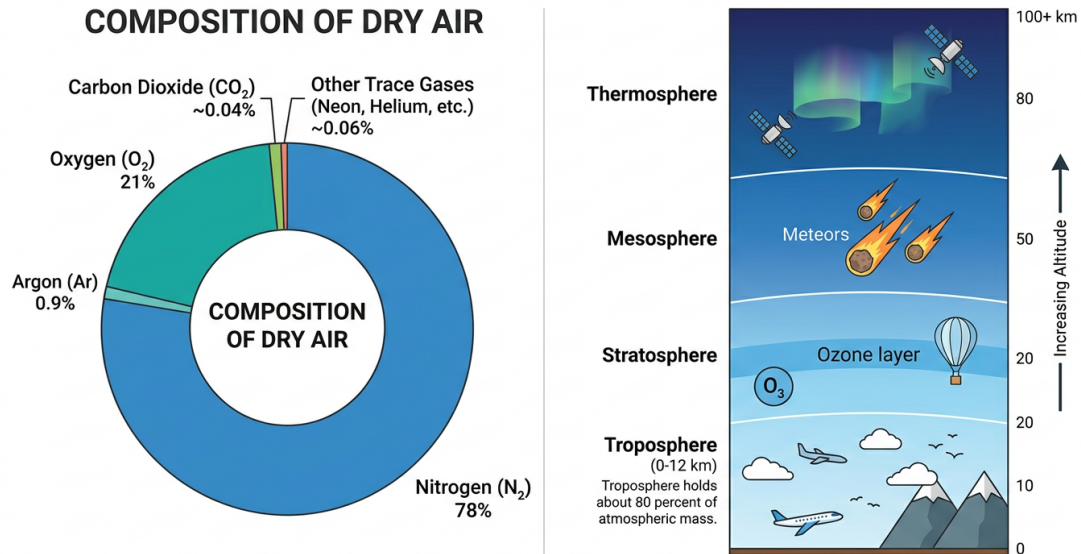
2. Air Pollution

Air Pollution

The **presence in the atmosphere of one or more air contaminants** in such quantities and of such duration as is (or tends to be) **injurious to human health or welfare, animal or plant life, or property**, or that would unreasonably interfere with the enjoyment of life or property.

Extended Definition

Presence in the outdoor air of **one or more contaminants, or combinations of them**, in such quantities and durations as may be (or tend to be) **injurious to human, animal or plant life, or property**, or which unreasonably interferes with the comfortable



Composition of air and the layers of the atmosphere.

enjoyment of life or property, or the conduct of business.

2.1 Forms of Air Pollutants

- Air pollutants come in the form of:
 - Gases**,
 - finely divided **solid aerosols**, and
 - finely divided **liquid aerosols**.
- Aerosols** = “any solid or liquid particles suspended in the air”.

2.2 Air Pollutant — Key Facts

- A pollutant can be **solid** (large or sub-molecular), **liquid**, or **gas**.
- It may originate from a **natural** or an **anthropogenic** (human-made) source — or both.
- Anthropogenic sources have changed the composition of global air by **less than 0.01%**.

2.2.1 Examples of Natural Air Pollution

- Ash
- Salt particles
- Pollen and spores
- Smoke
- Windblown dust

3. Sources and Classification of Air Pollutants

Air pollutants can be classified in several ways.

3.1 Classification 1 — According to Origin

3.1.1 A. Primary Pollutants

- Pollutants **emitted directly from identifiable sources**.
- Produced by **natural events** (e.g. dust storms and volcanic eruptions) and **human activities** (e.g. emissions from vehicles and industries).
- **Examples:** smoke, dust, oxides of sulphur and nitrogen, hydrocarbons, particulate matter.

3.1.2 B. Secondary Pollutants

- Pollutants **formed in the atmosphere** through chemical interactions between **primary pollutants** and **atmospheric constituents**.
- They are **not directly emitted** — they form through reactions.
- **Examples:** sulphur trioxide, ozone, ketones, sulphuric acid, nitric acid, carbonic acid.

Simple Understanding

Primary = directly released into the air. **Secondary** = formed in the air by reactions.

3.2 Classification 2 — According to State of Matter

3.2.1 A. Gaseous Air Pollutants

- Exist in a **gaseous state** at normal temperature and pressure.
- Examples: **carbon dioxide, nitrogen dioxide, sulphur oxides**.

3.2.2 B. Particulate Air Pollutants

- **Not gaseous** — they are suspended droplets, solid particles, or mixtures of the two.
- Examples: **dust, soot, smoke, fumes**.

3.3 Classification 3 — According to Sources

3.3.1 A. Natural Sources

- | | |
|-------------------------------|---|
| • Volcanic eruptions | • Marsh gases |
| • Deflation of sand and dust | • Cosmic dust |
| • Forest / wild fires | • Pollen grains of flowers |
| • Sulphur springs | • Photochemical reactions |
| • Natural geysers | • Soil debris |
| • Organic and inorganic decay | • Wild animals (decomposition gases, waste) |
| • Vegetative decay | |

3.3.2 B. Man-Made (Anthropogenic) Sources

- Industries and factories
- Urban centres
- Aircraft
- Nuclear experiments
- Automobiles
- Agriculture
- Domestic burning of wood
- Burning of fossil fuels
- Deforestation
- Mining
- Waste treatment plants
- Power plants

3.4 Classification 4 — According to Location of Emission Source

3.4.1 A. Stationary Sources

- An emission source that **does not move** (e.g. power plants, chemical and manufacturing industries).
- Often defined as **large emitters** releasing relatively consistent quantities of pollutants.

i. Point Source:

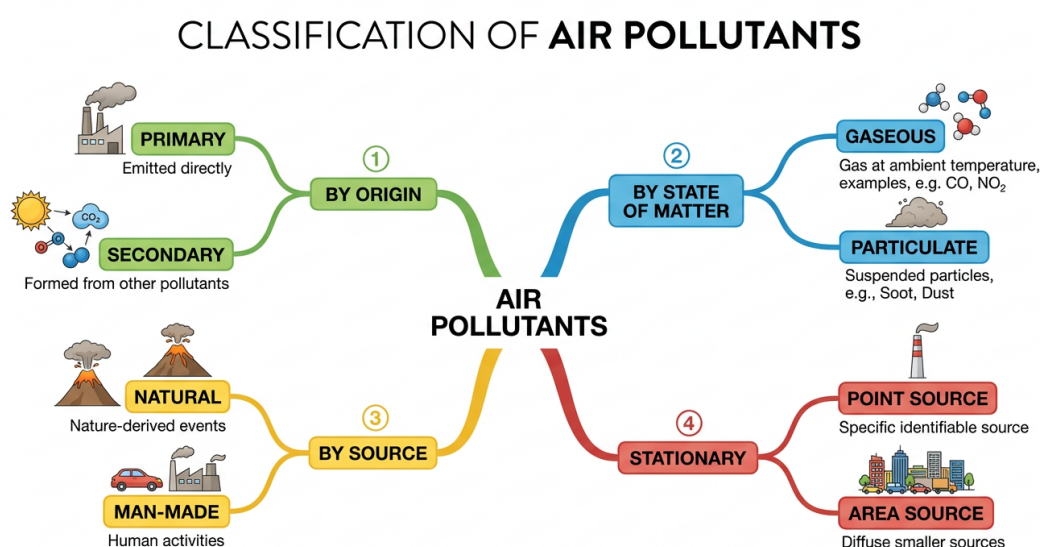
- **Industrial and non-industrial stationary equipment** or processes that are significant sources of air pollution.
- A facility is considered to have **significant emissions** if it emits **one ton or more in a calendar year**.

ii. Area Source:

- **Many smaller stationary sources** located together.
- Individual emissions may be **low**, but **collective emissions can be significant**.
- Typically emit **less than 25 tons per year** of any combination of hazardous air pollutants, or **less than 10 tons per year** of any single hazardous air pollutant.
- **Examples:** locomotives operating within a railyard; multiple flue gas stacks within a single plant; open burning and forest fires; evaporation losses from large spills of volatile liquids; piping leaks, industrial wastewater treatment ponds, rock and quarry operations, tank farms.

3.5 Factors that Affect Air Pollution

Factor	Explanation
Emissions	Traffic, industrial, and domestic emissions are major contributors
Geography (terrain)	Mountains and valleys can trap pollutants in certain areas
Weather conditions	Rain washes away pollutants; winds disperse them; humidity affects reactions
Season	Winter inversions can trap pollution; summer increases photochemical smog
Time of day	Morning and evening rush hours have higher traffic emissions
Population density	More people = more emissions, vehicles, and industrial activity
Indoor vs outdoor	Indoor air can be more polluted than outdoor air due to confined spaces



Classification of air pollutants.

4. National Ambient Air Quality Standards (NAAQS)

4.1 What is Ambient Air Quality?

- **Ambient air quality** = the condition or quality of the outdoor air.
- **NAAQS** = standards for ambient air quality set with reference to various identified pollutants.
- They are **notified by the CPCB** (Central Pollution Control Board) under the **Air (Prevention and Control of Pollution) Act, 1981**.

4.2 Major Objectives of NAAQS

- (i) To indicate the **necessary air-quality levels** and appropriate margins required to ensure the **protection of vegetation, health, and property**.
- (ii) To provide a **uniform yardstick** for the **assessment of air quality at the national level**.
- (iii) To **indicate the extent and need** of the monitoring programme.

4.3 NAAQS Standard Values (2009) — Key Pollutants

Pollutant	Time Period	Industrial / Residential / Rural	Ecologically Sensitive
SO ₂	Annual	50 µg/m ³	20 µg/m ³
SO ₂	24 Hours	80 µg/m ³	80 µg/m ³
NO ₂	Annual	40 µg/m ³	30 µg/m ³
NO ₂	24 Hours	80 µg/m ³	80 µg/m ³
PM ₁₀	Annual	60 µg/m ³	60 µg/m ³
PM ₁₀	24 Hours	100 µg/m ³	100 µg/m ³
PM _{2.5}	Annual	40 µg/m ³	40 µg/m ³
PM _{2.5}	24 Hours	60 µg/m ³	60 µg/m ³
Ozone (O ₃)	8 Hours	100 µg/m ³	100 µg/m ³
Ozone (O ₃)	1 Hour	180 µg/m ³	180 µg/m ³
Lead (Pb)	Annual	0.50 µg/m ³	0.50 µg/m ³
Carbon Monoxide (CO)	8 Hours	02 mg/m ³	02 mg/m ³
Ammonia (NH ₃)	Annual	100 µg/m ³	100 µg/m ³
Benzene (C ₆ H ₆)	Annual	05 µg/m ³	05 µg/m ³
Arsenic (As)	Annual	06 ng/m ³	06 ng/m ³
Nickel (Ni)	Annual	20 ng/m ³	20 ng/m ³

Important Notes on the Standards

- **Annual standards** = the annual arithmetic mean of a **minimum of 104 measurements** in a year at a site, taken **twice a week** at uniform 24-hourly intervals.
- The values must be complied with **98% of the time** in a year.
- There is a **2% chance of exceeding the limits**, but **not on two consecutive days** of monitoring.
- If results exceed the limits on **two consecutive days**, it is considered adequate reason to begin **regular/continuous monitoring** and further investigation.

5. Air Quality Index (AQI)

5.1 Background

- The AQI was launched by the **Prime Minister in April 2015**.
- It initially started with **14 cities**, and has now been extended to **71 cities in 17 states**.

5.2 What is the AQI?

- The AQI is a tool for the **effective communication of air-quality status to people**.
- It transforms **complex air-quality data** for various pollutants into a single, easy-to-understand form:
 - A **single number** (the index value),
 - a **nomenclature** (category name), and
 - a **colour** (colour coding).

5.3 How the AQI Works

- Sub-indices and health breakpoints are evolved for **eight pollutants: PM₁₀, PM_{2.5}, NO₂, SO₂, CO, O₃, NH₃, and Lead (Pb)**.
- For these, **short-term (up to 24-hour)** National Ambient Air Quality Standards are prescribed.
- Based on measured ambient concentrations, a **sub-index is calculated** as a linear function of concentration.
- **The worst sub-index determines the overall AQI.**

Example

The sub-index for PM_{2.5} will be **51** at a concentration of 31 µg/m³, **75** at 45 µg/m³, and **100** at 60 µg/m³.

5.4 AQI Categories and Health Breakpoints

AQI Range	Category	Associated Health Impacts
0–50	Good	Minimal impact
51–100	Satisfactory	May cause minor breathing discomfort to sensitive people
101–200	Moderate	May cause breathing discomfort to people with lung disease (e.g. asthma), and discomfort to people with heart disease, children, and older adults
201–300	Poor	May cause breathing discomfort to people on prolonged exposure, and discomfort to people with heart disease on short exposure
301–400	Very Poor	May cause respiratory illness on prolonged exposure; effect more pronounced in people with lung and heart diseases
401–500	Severe	May cause respiratory effects even in healthy people, and serious impacts on those with lung/heart disease; effects felt even during light physical activity

Six AQI Categories

Good, Satisfactory, Moderately Polluted, Poor, Very Poor, and Severe.

Air Quality Index AQI



The Air Quality Index (AQI) colour scale.

6. Effects of Air Pollution on Human Health

6.1 1. Tobacco Smoke

- Generates a **wide range of harmful chemicals** and is a **major cause of cancer**.
- Affects even **passive smokers** (non-smokers in the vicinity).
- Effects range from a burning sensation in the eyes or nose, throat irritation, bronchitis, and severe asthma, to a **decrease in lung function** and **cancer**.

6.2 2. Biological Pollutants

- Mostly **allergens**.
- Can cause **asthma, hay fever**, and other allergic diseases.

6.3 3. Volatile Organic Compounds (VOCs)

- Can cause **irritation of the eyes, nose, and throat**.
- In severe cases: **headaches, nausea, loss of coordination**.
- **Long-term exposure** is suspected to damage the **liver and other body parts**.

6.4 4. Formaldehyde

- Exposure causes **irritation of the eyes and nose**.
- May cause **allergies** in some people.

6.5 5. Lead (Pb)

- **Prolonged exposure** damages the **nervous system**, causes **digestive problems**, and in some cases **cancer**.
- **Especially hazardous to small children**.

6.6 6. Radon

- A **radioactive gas** that can **accumulate inside houses**.
- **Originates from rocks and soil** beneath the house.
- Its indoor level is influenced by outdoor air and other indoor gases.
- Exposure increases the **risk of lung cancer**.

6.7 7. Ground-Level Ozone (O₃)

- Makes the **eyes itch and burn**.
- Associated with an **increase in respiratory disorders** such as asthma.
- **Lowers resistance** to colds and pneumonia.

6.8 8. Oxides of Nitrogen (NO_x)

- Can make **children susceptible to respiratory diseases** in winter.
- Contributes to **smog formation**.

6.9 9. Carbon Monoxide (CO)

- **CO combines with haemoglobin**, lessening the amount of oxygen entering the blood through the lungs.
- This binding changes the function of affected organs: the **brain, cardiovascular system, and developing foetus**.
- Effects: **impairs concentration, slows reflexes**, and makes a person **confused and sleepy**.

6.10 10. Sulphur Dioxide (SO₂)

- Present in air due to the **combustion of fossil fuels**.
- Can **oxidise to form sulphuric-acid mist**.
- Leads to **diseases of the lung** and other disorders — **wheezing, shortness of breath**.
- Long-term effects are hard to isolate, as SO₂ exposure often occurs together with SPM.

6.11 11. Suspended Particulate Matter (SPM)

- Consists of **dust, fumes, mist, and smoke**.
- Main chemical component of concern: **Lead (Pb)**; other components include **nickel, arsenic, diesel exhaust particles**.
- When breathed in, it **lodges in lung tissue**, causing **lung damage and respiratory problems**.

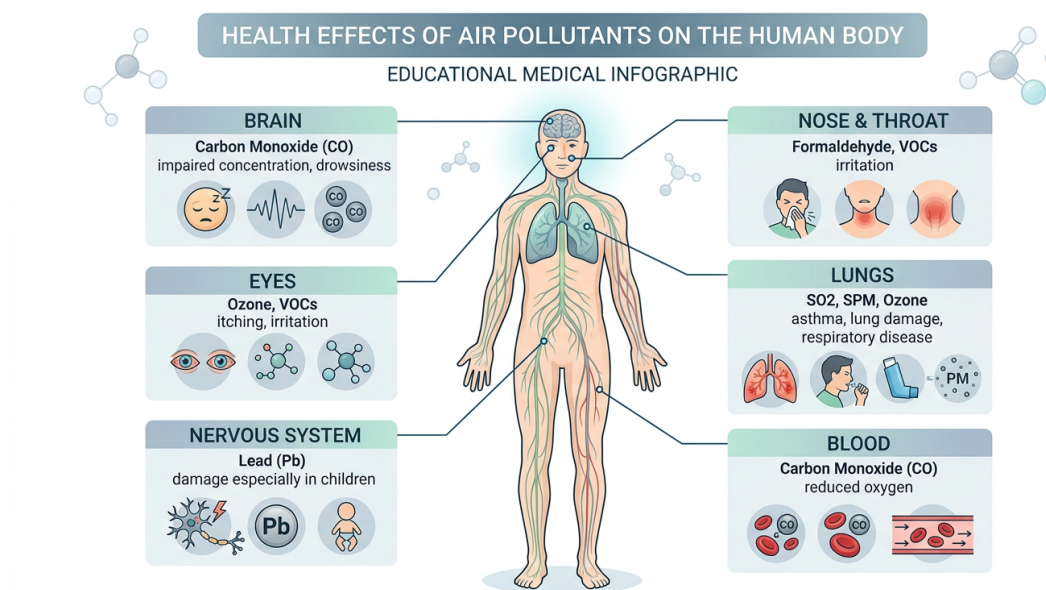
Why SPM is a Major Pollutant

- a) It affects **more people globally** than any other pollutant on a continuing basis.
- b) There is **more monitoring data** available on it than on any other pollutant.
- c) More **epidemiological evidence** has been collected on exposure to it than to any other pollutant.

7. Economic Effects of Air Pollution

7.1 Overview

- Air pollution affects **human health, agriculture**, and leads to a range of other impacts.
- These impacts are projected to become **much more severe** in the coming decades.
- Rising emissions are projected to lead to **higher concentrations of PM_{2.5} and ground-level ozone**.
- In several regions, PM_{2.5} and ozone concentrations are **already above WHO guidelines**.



Air pollutants and their effects on the human body.

7.2 Economic Statistics

Parameter	Current Value	Projected (2060)
Global economic costs	—	1% of global GDP
Healthcare costs	USD 21 billion (2015)	USD 176 billion
Lost working days	~1.2 billion/year	3.7 billion/year
Premature deaths	~3 million (2010)	6–9 million/year
Global welfare costs (premature deaths)	USD 3 trillion (2015)	USD 18–25 trillion
Welfare costs (pain & suffering from illness)	USD 300 billion (2015)	USD 2.2 trillion

Key Point

A large number of deaths occur in densely populated regions with high concentrations of PM_{2.5} and ozone — especially **China and India** — and in regions with ageing populations, such as **China and Eastern Europe**.

7.3 Effects on Materials and Property

Material	Effect of Air Pollution
Building materials	Corroded/disfigured; SO ₂ /SO ₃ with moisture react with limestone (CaCO ₃) to form calcium sulphate and gypsum; stone converts to soluble bicarbonate and leaches away; needs extra cleaning
Paints	High-pollution areas need more frequent painting; pollutants damage the protective coating and expose the surface
Textiles	Sulphur oxides deteriorate natural and some synthetic fibres; classic damage is the deterioration of nylon hose
Rubber	Side walls of tyres and electrical insulation are damaged — caused by ozone
Leather	Sulphur dioxide makes leather lose strength and ultimately disintegrate
Paper	Metallic impurities accelerate conversion of absorbed SO ₂ to sulphuric acid in moisture
Glass & ceramics	Long exposure changes surface appearance; moisture + acidic pollution degrade the surface
Electronics industry	Pollutants form a thin insulating film on contacts, causing open circuits and malfunction
Vegetation	Damage to vegetation is a very important aspect of economic loss

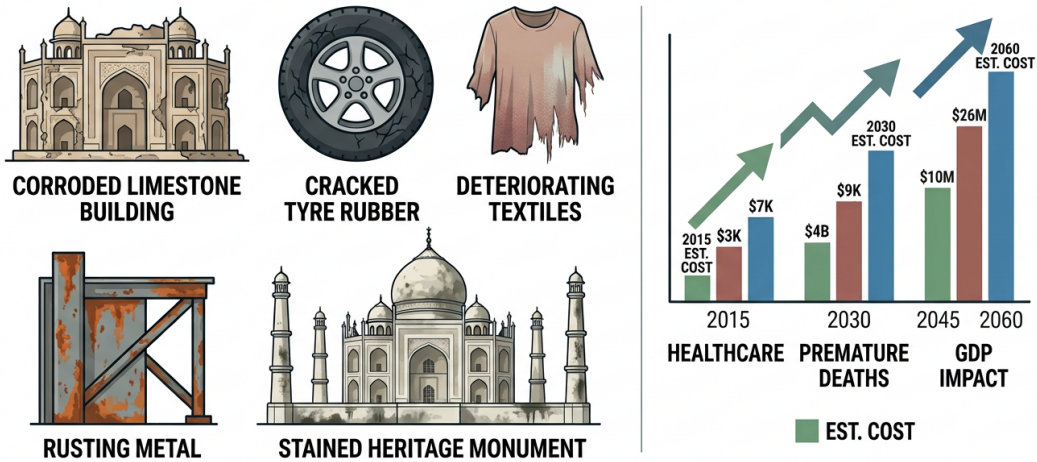
Intangible Loss — Heritage Monuments

Air pollution damages **art objects and heritage monuments** across the world. Examples of affected sites include the **Taj Mahal (Agra, India)**, the **Sri Channakeshava Temple (Belur, India)**, and the **Parthenon (Athens, Greece)**.

7.4 Policy Implication

- **Policies to limit air-pollution emissions** would improve air quality, reduce the risk of very severe health impacts, and generate considerable **climate co-benefits**.
- Other effects of pollution include corrosion of metals, soiling and eroding of building surfaces, fading of dyed materials, cracking of rubber, spoiling of vegetation, effects on animals, and interference with production and services.

ECONOMIC EFFECTS OF AIR POLLUTION



Economic damage caused by air pollution.

8. Control of Air Pollution by Equipment

Air Pollution Control Devices

A series of devices that work to **prevent a variety of pollutants** (both gaseous and solid) from entering the atmosphere — primarily from **industrial smokestacks**.

They fall into **two broad categories**:

1. Devices that control **particulate matter**.
2. Devices that control **acidic gas emissions**.

Note

Extraction methods differ for each type of pollutant. Although complex, these devices have proven effective — overall emission levels have dropped with their implementation.

8.1 A. Control of Particulate Matter

Specific machinery removes particulate matter from flue gases using **physical means of separation** (since the particles are large enough to be “caught”).

8.1.1 1. Electrostatic Precipitators (ESP)

- **Definition:** a filter that uses **static electricity** to remove soot and ash from exhaust fumes before they leave the smokestacks.
- **Working principle:** unburned carbon particles are pulled out of the smoke by static electricity — **negative electrodes** are metal wires, **positive electrodes** are large plates; charged particles are attracted to the oppositely charged plates, leaving clean hot air to escape.

- **Why important:** vital to remove unreacted carbon, which can **damage buildings** and **harm respiratory health**.

8.1.2 2. Cyclone Separators

- **Definition:** a device that uses the **principle of inertia** to remove particulate matter from flue gases.
- **Working principle:** dirty flue gas enters a chamber containing a **vortex** (like a tornado). Due to the inertia difference, **gas particles move up** the cylinder while **larger particles hit the wall and drop down**. **Cleaned flue gas exits from the top**.

8.1.3 3. Fabric Filters

- **Definition:** a fairly simple method to remove **dust from flue gases**.
- **Working principle:** fabric (generally **felt**, not woven cloth, which would let dust through) is placed so flue gases must **pass through it**; **dust particles are trapped in the cloth**. Can also remove **acidic gases** if basic compounds are used.

8.2 B. Gas Control

More intense chemical methods are needed to separate polluting gases from flue gas — important because many **acidic gases contribute to acid rain**.

8.2.1 1. Scrubbers

- **Definition:** a system used to **remove harmful materials from industrial exhaust gases** before release. When used specifically to remove **SO_x**, it is called “**Flue Gas Desulfurization**”.

Type	Working
Wet Scrubbers	Remove harmful components of flue gas by spraying a liquid substance through the gas
Dry Scrubbers	Remove harmful components by introducing a solid substance (usually powdered) into the gas

- **Purpose:** by removing acidic gases before release, scrubbers help **prevent the formation of acid rain**.

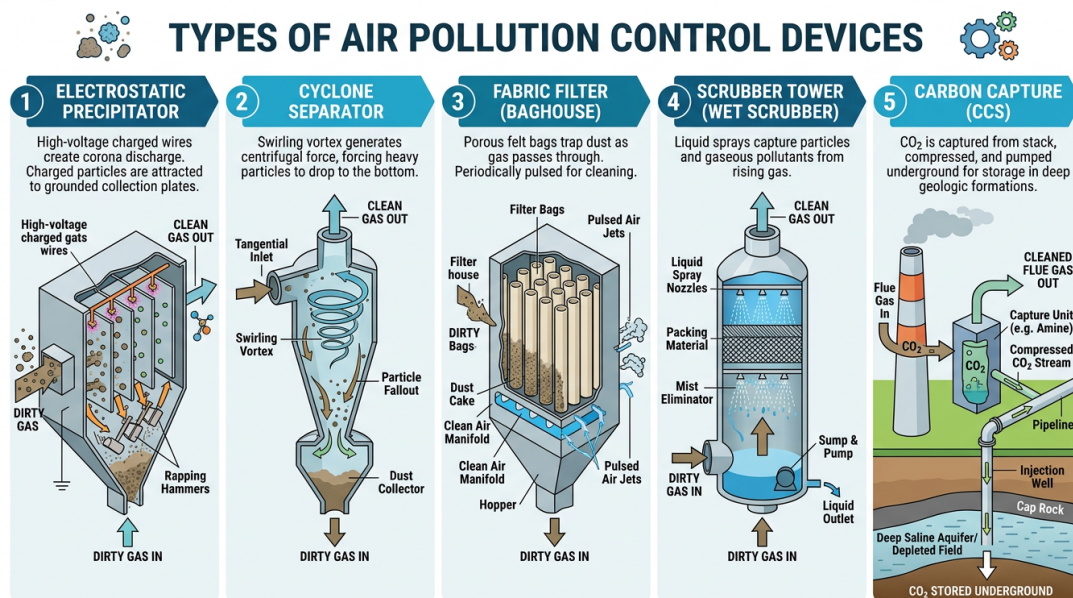
8.2.2 2. Incineration

- **Definition:** used to **convert VOC emissions into carbon dioxide and water** through combustion.
- **Working principle:** takes place in an **afterburner** built to create the conditions for **complete combustion** — sufficient burn time and high temperature; the gas must be **mixed** to ensure complete combustion.
- **Types:** bubbling-bed technology and circulating-bed technology (fluidised-bed incinerator).

8.2.3 3. Carbon Capture

- **Definition:** **Carbon Capture and Storage (CCS)** — the process of capturing **CO₂** and storing it below ground.

- **Working principle:** CO₂ is captured and stored **underground** (pumped into geologic layers) or in **forests and oceans**, preventing it from entering the atmosphere.
- **Current status:** rarely used at present, but widely discussed as a way to **limit greenhouse-gas emissions** driving climate change.



Air pollution control equipment.

9. Smoke and Its Control

9.1 What is Smoke?

- Smoke is a **collection of tiny solid, liquid, and gas particles**.
- It is a type of **aerosol** released during the **combustion of materials**.
- Main components: **carbon (soot), ash, and volatile organic compounds**.

9.2 Control of Smoke

- Smoke control is achieved through the **air-pollution control devices** described above (ESP, cyclone separators, scrubbers, fabric filters).
- Additional measures:
 - Using **cleaner fuels** (natural gas instead of coal).
 - **Better combustion technology** for complete burning.
 - **Catalytic converters** in vehicles.
 - **Regular inspection** and maintenance of industrial sources.

10. Ozone Depletion

10.1 What is the Ozone Layer?

- The **ozone layer** is like an **umbrella**, located about **24 km (15 miles)** above the Earth's surface.
- It is an **essential component of the stratosphere**.
- **Function:** it absorbs **short-wavelength ultraviolet (UV) radiation** from the sun, heating the gases of the stratosphere in the process.
- This is why **temperature rises with increasing altitude** in the stratosphere.
- The ozone layer is the reason **life was able to move out of the oceans** and onto land, evolving into today's diverse biosphere.

10.2 Ozone Layer — Key Facts

Fact	Data
World Ozone Day	Celebrated on 16 September every year
Unit of measurement	Dobson Units (DU) — named after G.M.B. Dobson
1 Dobson Unit	= 0.01 mm thickness of stratospheric ozone
Average ozone thickness	300 DU
Ozone-hole threshold	When it falls below 200 DU
Thickness variation	Thinnest around the equator, thickest near the poles

10.3 Causes of Ozone Depletion

10.3.1 Primary Cause

- Release of **Chlorofluorocarbons (CFCs)** and other **Ozone-Depleting Substances (ODS)**.
- CFCs were widely used as **refrigerants, insulating foams, and solvents**.

10.3.2 Mechanism of Ozone Destruction

Step 1: Although CFCs are heavier than air, they are eventually carried into the stratosphere — a process that can take **2 to 5 years**.

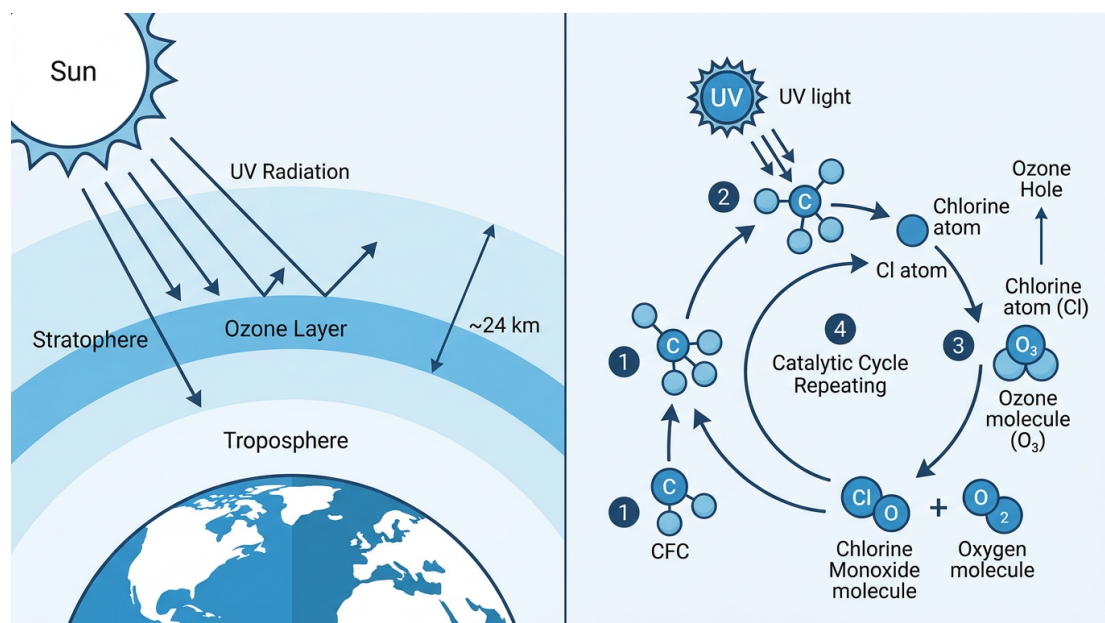
Step 2: In the stratosphere, **UV radiation** causes them to **break apart**.

Step 3: This releases **chlorine atoms**.

Step 4: Chlorine atoms **react with ozone (O₃)**, starting **chemical cycles of ozone destruction**.

Key Fact

One chlorine atom can break apart more than 1,00,000 (100,000) ozone molecules!



The ozone layer and how CFCs deplete it.

10.4 Effects of Ozone Depletion

10.4.1 Health Effects on Humans (from increased UV at the surface)

Health Effect	Explanation
Cataract	UV radiation damages the lens of the eye
Genetic mutation	UV rays can damage DNA
Constriction of blood vessels	UV exposure affects the circulatory system
Leukaemia	Blood cancer linked to UV radiation
Breast cancer	Linked to UV exposure
Skin cancer (basal & squamous cell)	Strongly linked to UVB exposure
Malignant melanoma	Less common but far more dangerous — lethal in about 15–20% of diagnosed cases
Snow blindness (photo-keratosis)	Inflammation of the cornea
Reduced resistance to infection	UV suppresses the immune system

10.4.2 Environmental Effects

- **Reduced crop yield** — UV damages agricultural crops.
- **Damage to aquaculture** — affects marine organisms and food chains.
- Increased **tropospheric ozone** (a ground-level health risk).
- Increased **Vitamin D synthesis** in sunlight (can be positive or negative).

Important Note

These impacts are **not due to direct exposure to ozone**, but due to the **UV rays** that reach the Earth's surface through the ozone holes.

10.5 Control Measures for Ozone Depletion

10.5.1 Historical Timeline

Year	Event
1979	Ozone hole over Antarctica first spotted
1957–1985	An 18-year study showed a 35% decrease in the ozone column over Antarctica
1978	Use of CFC propellants in spray cans banned in the US
1980s	Antarctic ozone hole appeared; CFCs more strongly linked to depletion
1987	Montreal Protocol signed — nations committed to reducing CFCs and other ODS
1995	CFC production banned in developed countries after 1995
Later	Ban extended to developing countries
2000	International agreement (139 nations) banning CFC production
~2050	Ozone layer expected to return to normal levels if ODS production stops

10.5.2 Montreal Protocol (1987)

- An **international agreement**, originally signed by **139 nations**; today **over 160 countries** have signed.
- Committed nations to **reduce the use of CFCs** and other ozone-depleting substances.
- Amended to **ban CFC production after 1995** in developed countries.

10.5.3 Recovery Forecast

- We **cannot make enough ozone** to replace what has been destroyed.
- But if we **stop producing ODS**, natural ozone-production reactions should **return the layer to normal by about 2050**.
- Delays in ending production could cause **additional damage** and **prolong recovery**.

10.5.4 Replacement Chemicals

- The control mechanism stresses **replacing banned CFCs** with **ammonia, steam, helium**, and other non-ozone-depleting alternatives.

11. Photochemical Changes

11.1 What are Photochemical Changes?

- Photochemical changes occur when **sunlight reacts with pollutants** in the atmosphere.

- UV radiation initiates **chemical reactions** between pollutants.
- This is how many **secondary pollutants** are formed.

11.2 Photochemical Smog

Photochemical Smog

A type of air pollution resulting from chemical reactions of sunlight with hydrocarbons and nitrogen oxides released by automobiles and industrial sources.

11.2.1 Key Reactions

- $\text{NO}_2 + \text{sunlight} \rightarrow \text{NO} + \text{O}$ (atomic oxygen).
- Atomic oxygen reacts with $\text{O}_2 \rightarrow$ forms **O_3 (ozone)** at ground level.
- Ground-level ozone + hydrocarbons $\rightarrow$ forms **PAN (Peroxyacetyl Nitrate)** and other harmful compounds.

11.2.2 Effects of Photochemical Smog

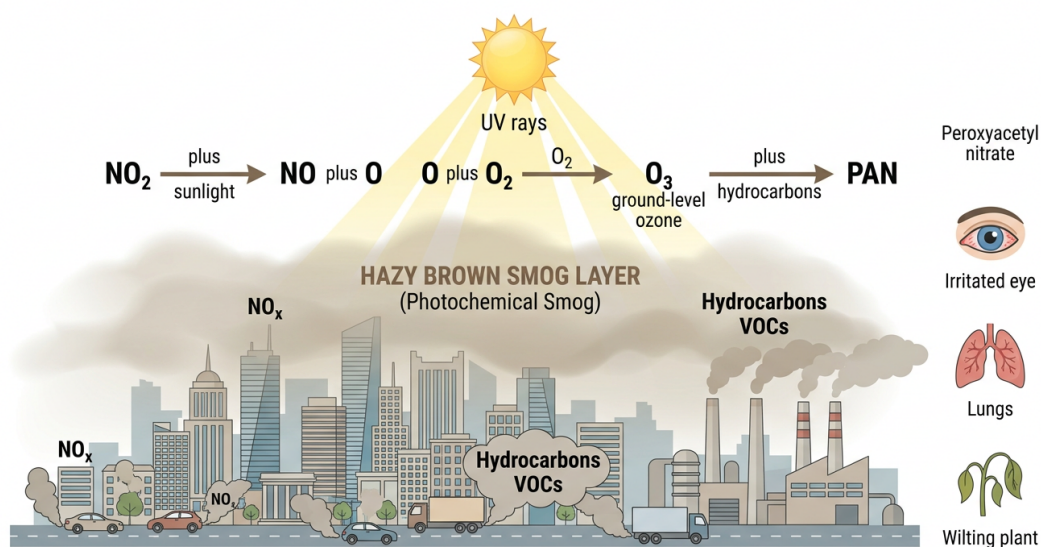
- Eye and throat irritation.
- Respiratory problems.
- Reduced visibility (**haze**).
- Damage to **vegetation and crops**.
- Aggravation of **asthma and lung diseases**.

11.3 Photochemical Changes in Ozone Depletion

- In the stratosphere, UV radiation causes **photolysis of CFCs**, releasing chlorine.
- $\text{Cl} + \text{O}_3 \rightarrow \text{ClO} + \text{O}_2$ (chlorine monoxide + oxygen).
- $\text{ClO} + \text{O} \rightarrow \text{Cl} + \text{O}_2$ (chlorine released again).
- This catalytic cycle **destroys ozone continuously** — one chlorine atom can destroy **100,000 ozone molecules**.

11.4 Summary of Photochemical Effects

Location	Process	Result
Troposphere (ground level)	Sunlight + NO_x + VOCs $\rightarrow$	Photochemical smog, ground-level ozone (harmful)
Stratosphere	UV + CFCs $\rightarrow$	Ozone depletion (harmful)



Formation of photochemical smog.

11. Summary Table — Module 3: Air

Topic	Key Concept
Air Composition	78% N_2 , 20.9% O_2 , 0.9% Ar, trace CO_2 and others
Troposphere	Lowest atmospheric layer; extends to 12 km; holds 80% of atmospheric mass
Air Pollution Definition	Presence of contaminants in air injurious to humans, animals, plants, or property
Air Pollutant Forms	Gases, solid aerosols, liquid aerosols
Classification by Origin	Primary (directly emitted) & Secondary (formed by reactions)
Classification by State	Gaseous (CO_2 , SO_2 , NO_2) & Particulate (dust, soot, smoke)
Classification by Source	Natural (volcanoes, dust storms) & Man-made (industries, vehicles, burning)
Emission Sources	Stationary (point source, area source)
NAAQS	Standards by CPCB under Air Act 1981; 12+ pollutants; reviewed 2009
AQI	Launched April 2015; 6 categories (Good to Severe); based on 8 pollutants
Health Effects	CO (less O_2 in blood), SO_2 (lung disease), Lead (nerve damage), SPM (lung damage), Ozone (eye/respiratory), Radon (lung cancer)

Topic	Key Concept
Economic Effects	1% of GDP by 2060; 6–9 million premature deaths; damage to materials, heritage, vegetation
Particulate Control	Electrostatic Precipitators, Cyclone Separators, Fabric Filters
Gas Control	Scrubbers (wet/dry), Incineration (afterburner), Carbon Capture (CCS)
Ozone Layer	At 24 km; absorbs UV; measured in Dobson Units; normal = 300 DU; hole below 200 DU
Ozone Depletion Cause	CFCs + UV → Cl atoms destroy O ₃ ; one Cl destroys 100,000 O ₃
Ozone Depletion Effects	Skin cancer, cataracts, genetic mutation, crop damage, snow blindness
Montreal Protocol	1987; 160+ countries; bans CFC production; recovery expected by 2050
Photochemical Changes	Sunlight + NO _x + VOCs = ozone + PAN + smog; UV + CFCs = stratospheric ozone loss

In a nutshell

These are the complete, detailed study notes of **Module 3: Air (BCV755B — Conservation of Natural Resources)**, covering every topic in the module — from the composition of air and air pollution, through classification, standards (NAAQS & AQI), health and economic effects, and control equipment, to ozone depletion and photochemical changes — without skipping anything.